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Heskett

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(54) **DEVICE AND METHODS FOR FREEZING LIQUIDS INTO A FLAT FORM**

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U.S.C. 154(b) by 186 days.

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(51) **Int. Cl.**
F25C 1/12 (2006.01)

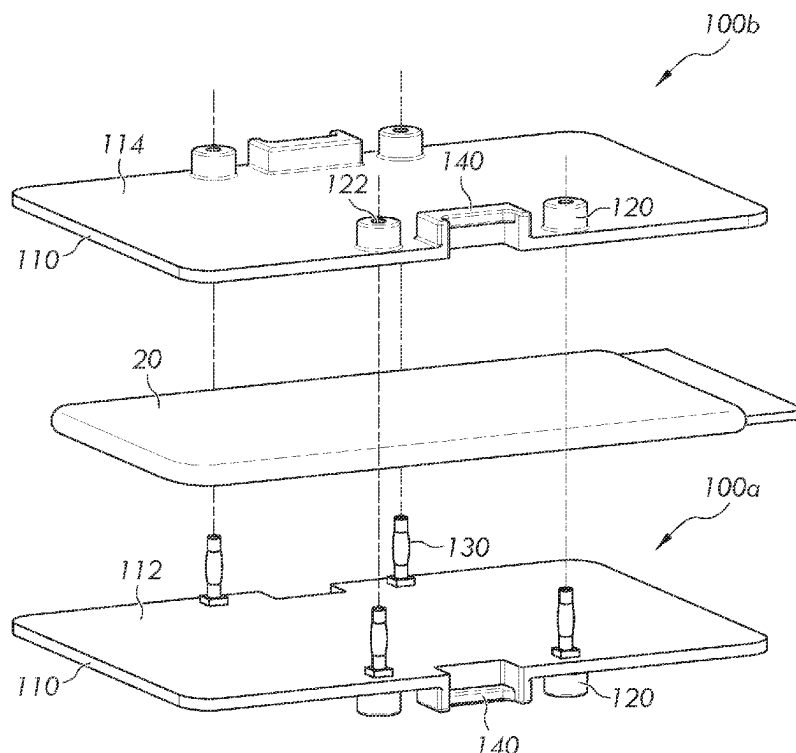
(57) **ABSTRACT**

(52) **U.S. Cl.**
CPC **F25C 1/12** (2013.01)

(58) **Field of Classification Search**
CPC F25C 1/22; F25C 1/12; F25D
2331/801–2331/8015; A61M 1/0277;
A61J 1/165; A23B 11/1455; A23B
2/8033; A23B 2/8055; A23B 4/068
See application file for complete search history.

A device for freezing liquid in a flat form has a first plate with a first portion of a latching mechanism and a second plate including a second portion of the latching mechanism. The latching mechanism includes a spring pin, a post and grommet, or a clip in various embodiments. The first plate and second plate can include a plurality of symmetrically formed tabs extending outward from the plates with the latching mechanism formed on the tabs. The first plate and second plate may include a plurality of symmetrically formed latching mechanisms.

20 Claims, 16 Drawing Sheets



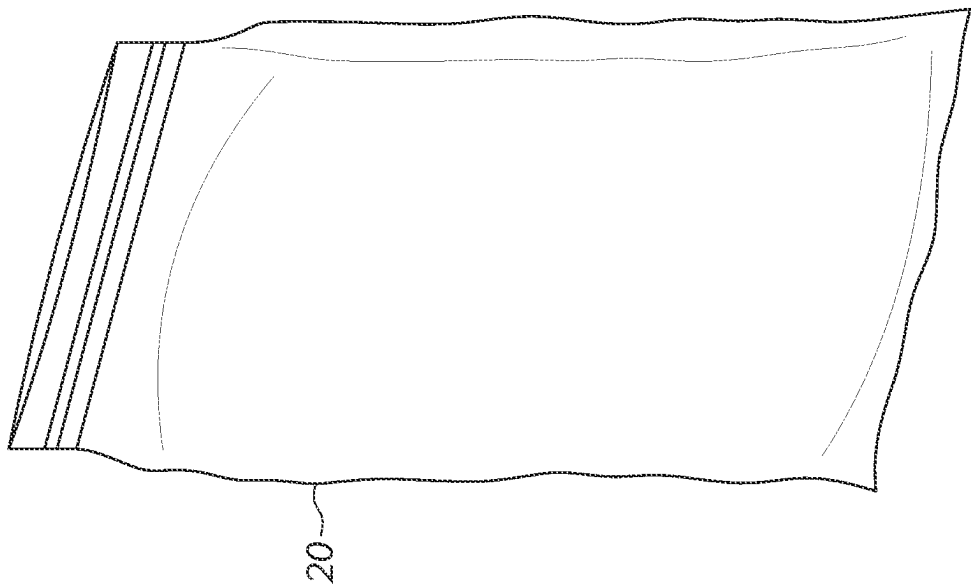


FIG. 1a
(PRIOR ART)

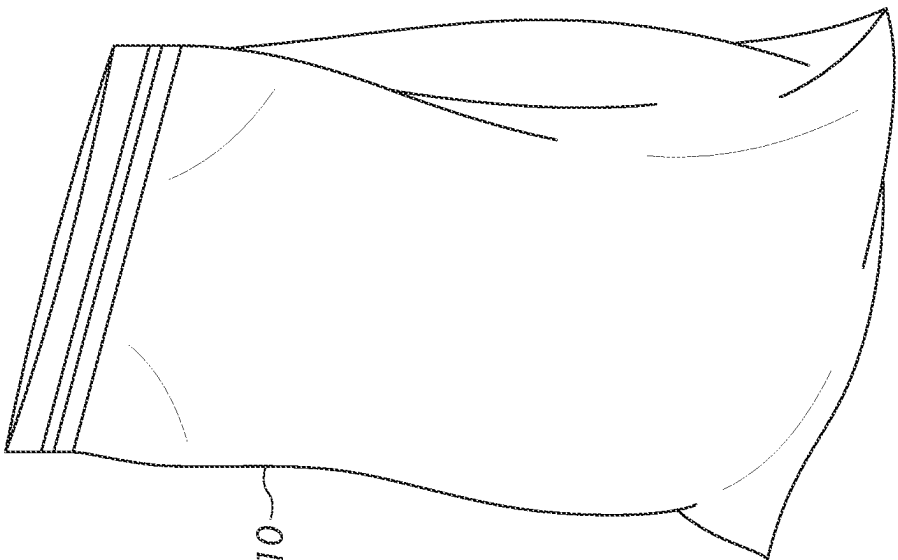


FIG. 1b
(PRIOR ART)

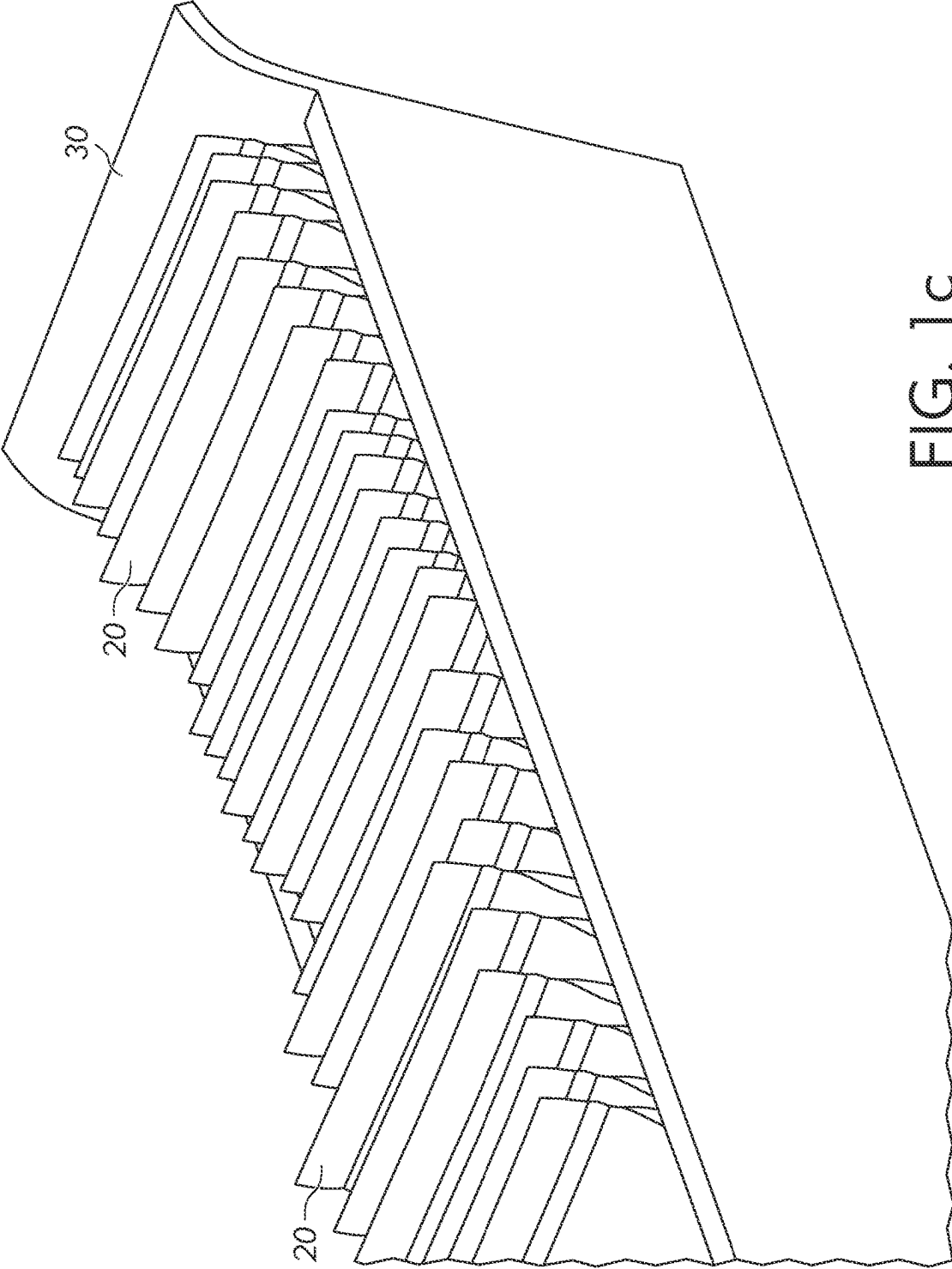


FIG. 1C
(PRIOR ART)

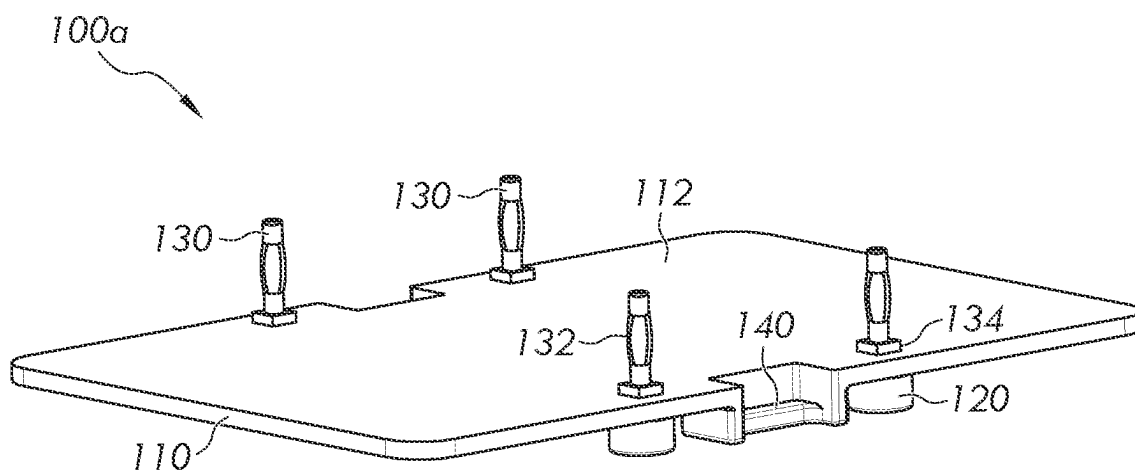


FIG. 2a

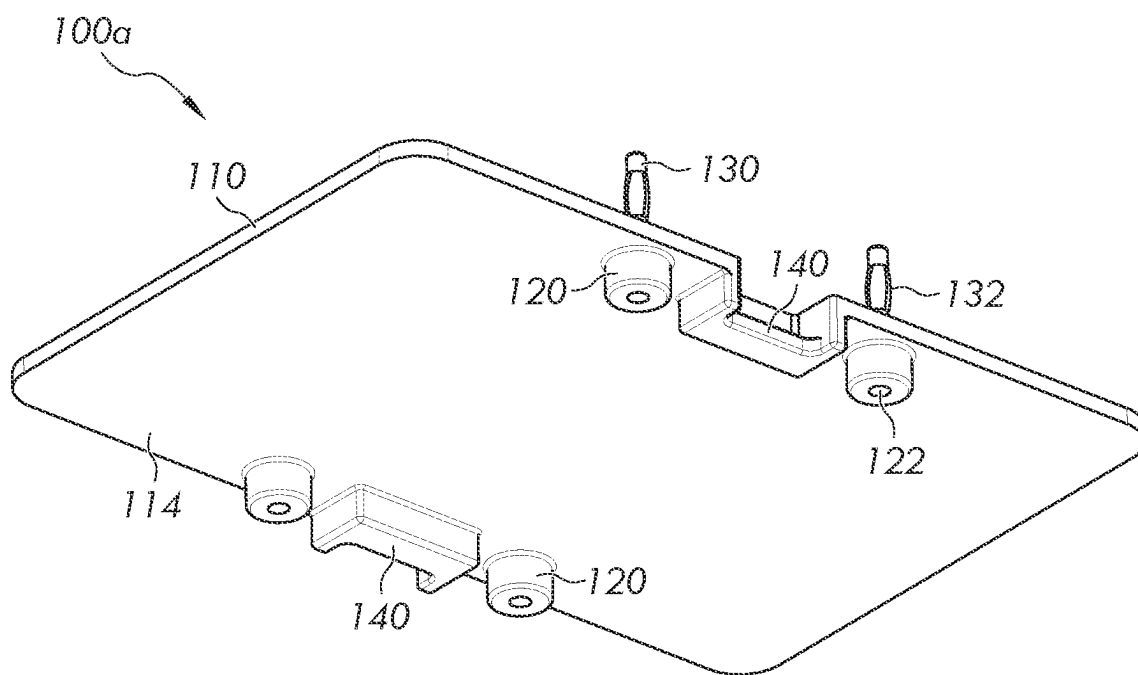


FIG. 2b

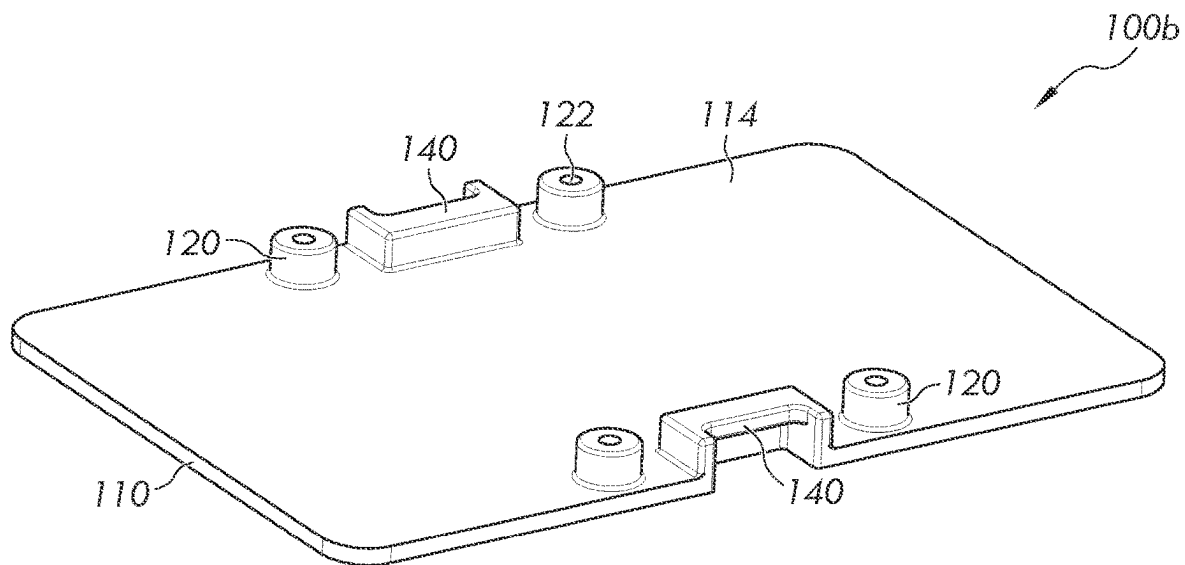


FIG. 2c

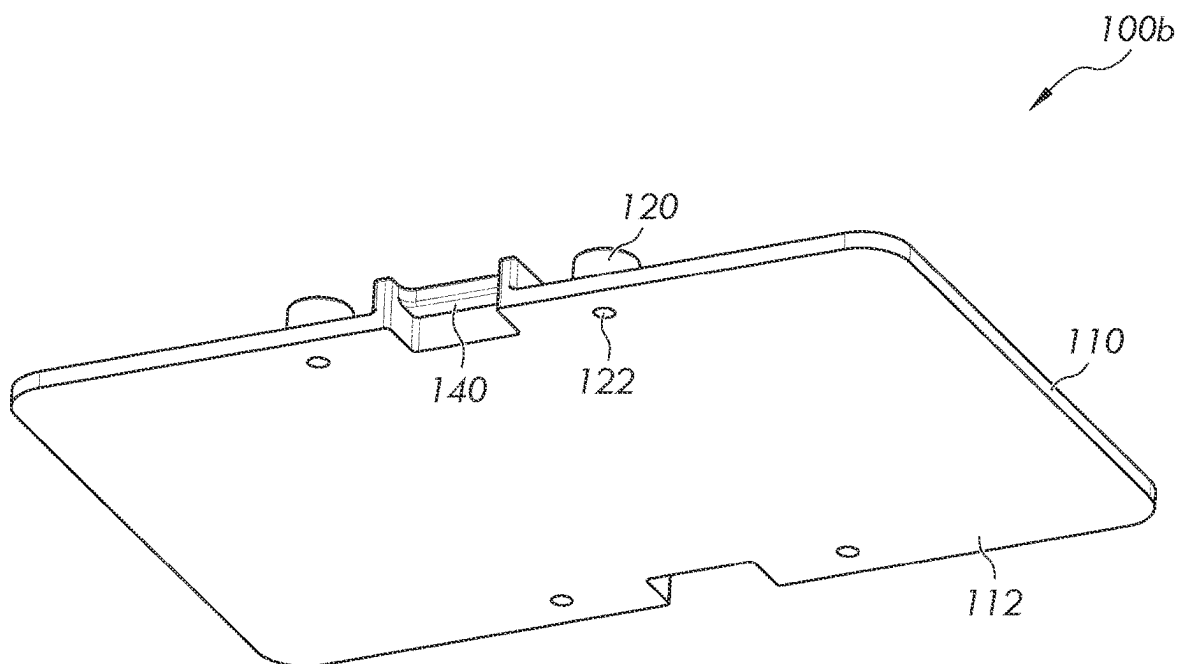


FIG. 2d

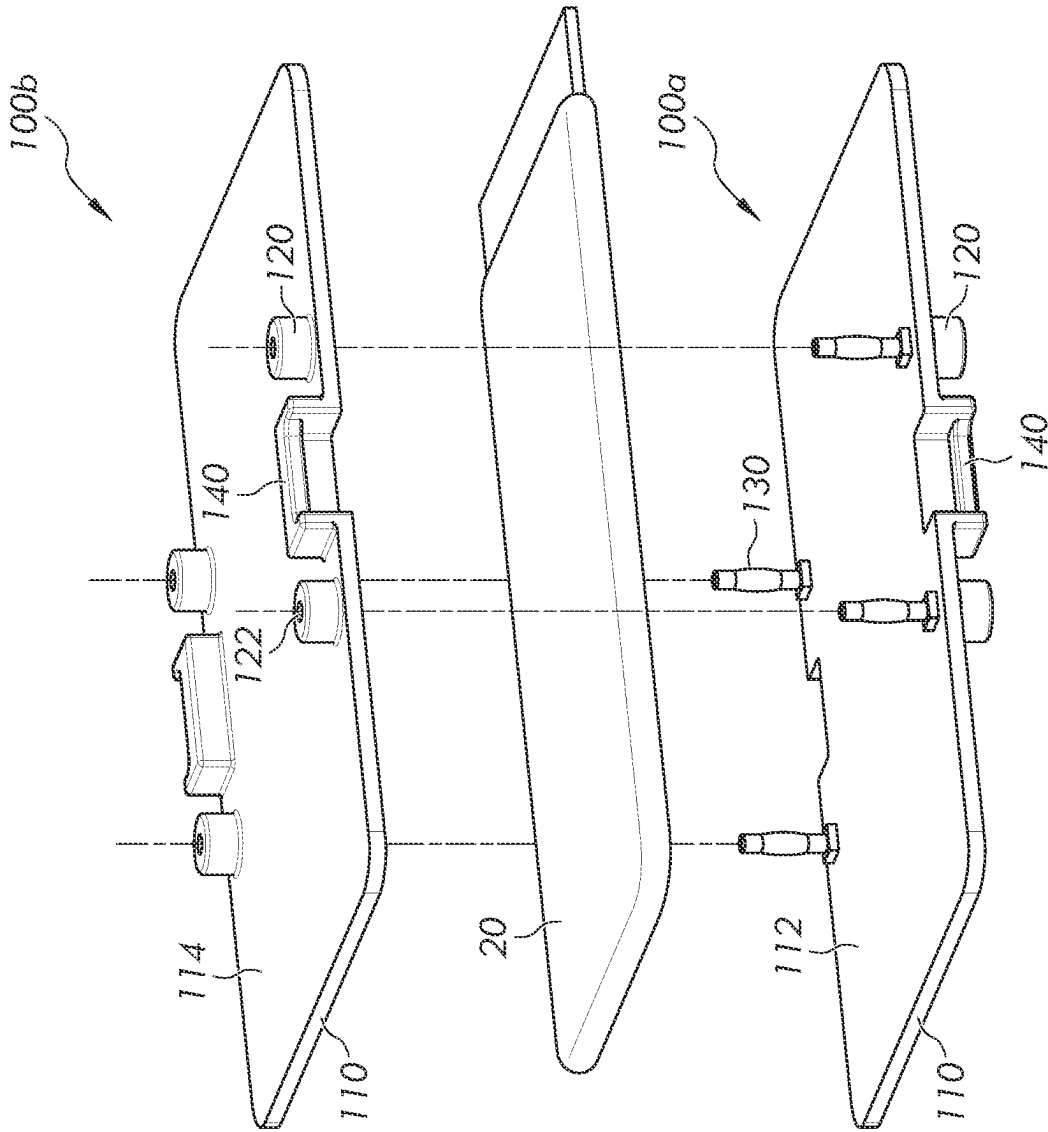


FIG. 2f

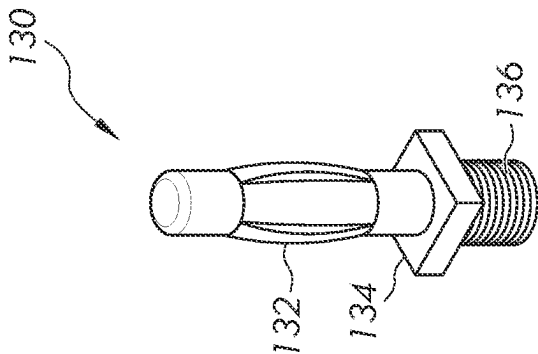


FIG. 2e

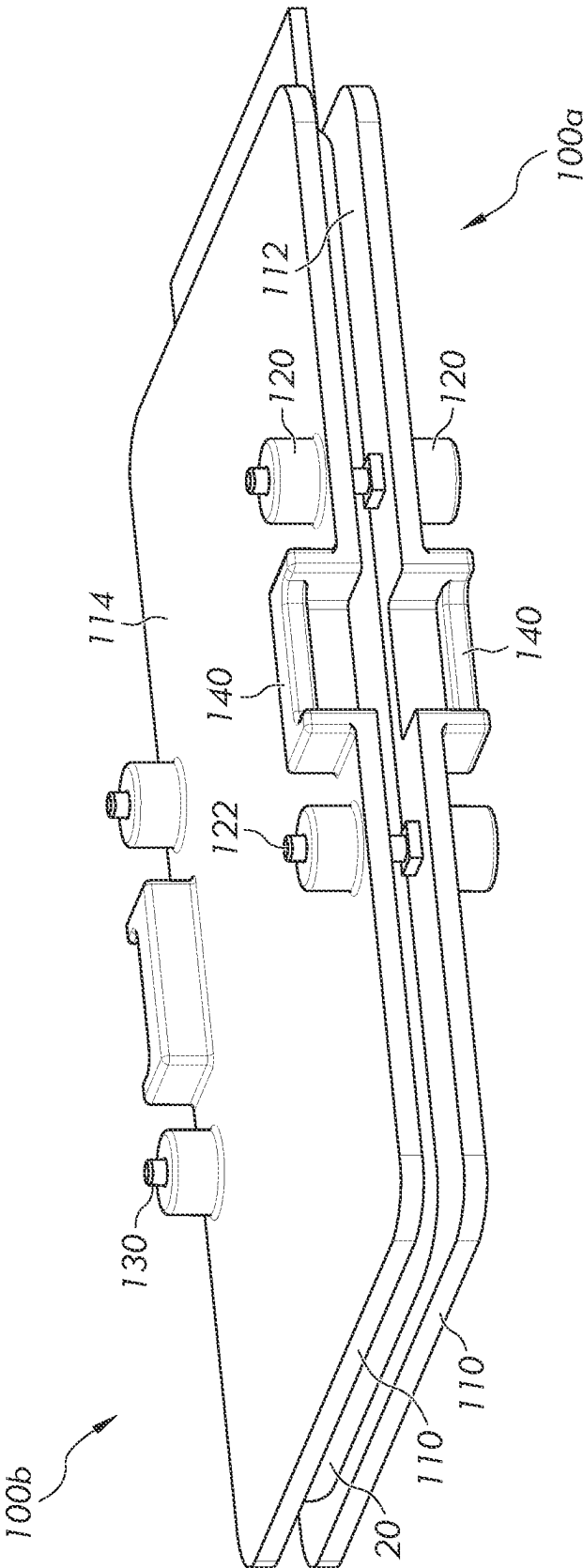


FIG. 2g

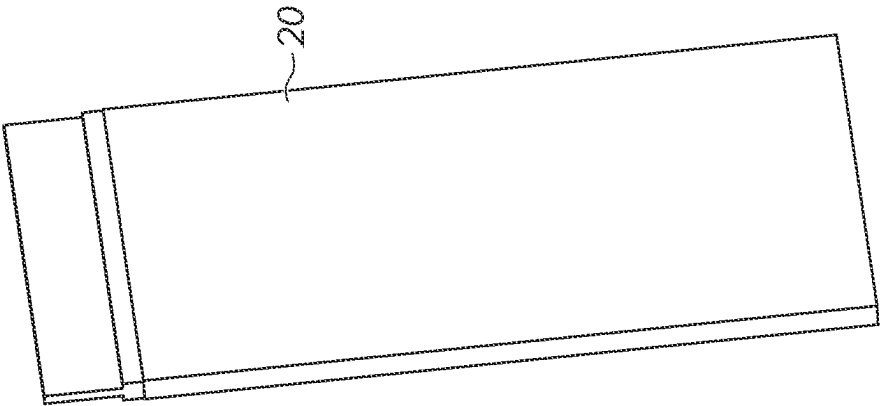


FIG. 2h

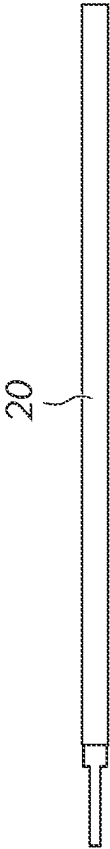


FIG. 2i

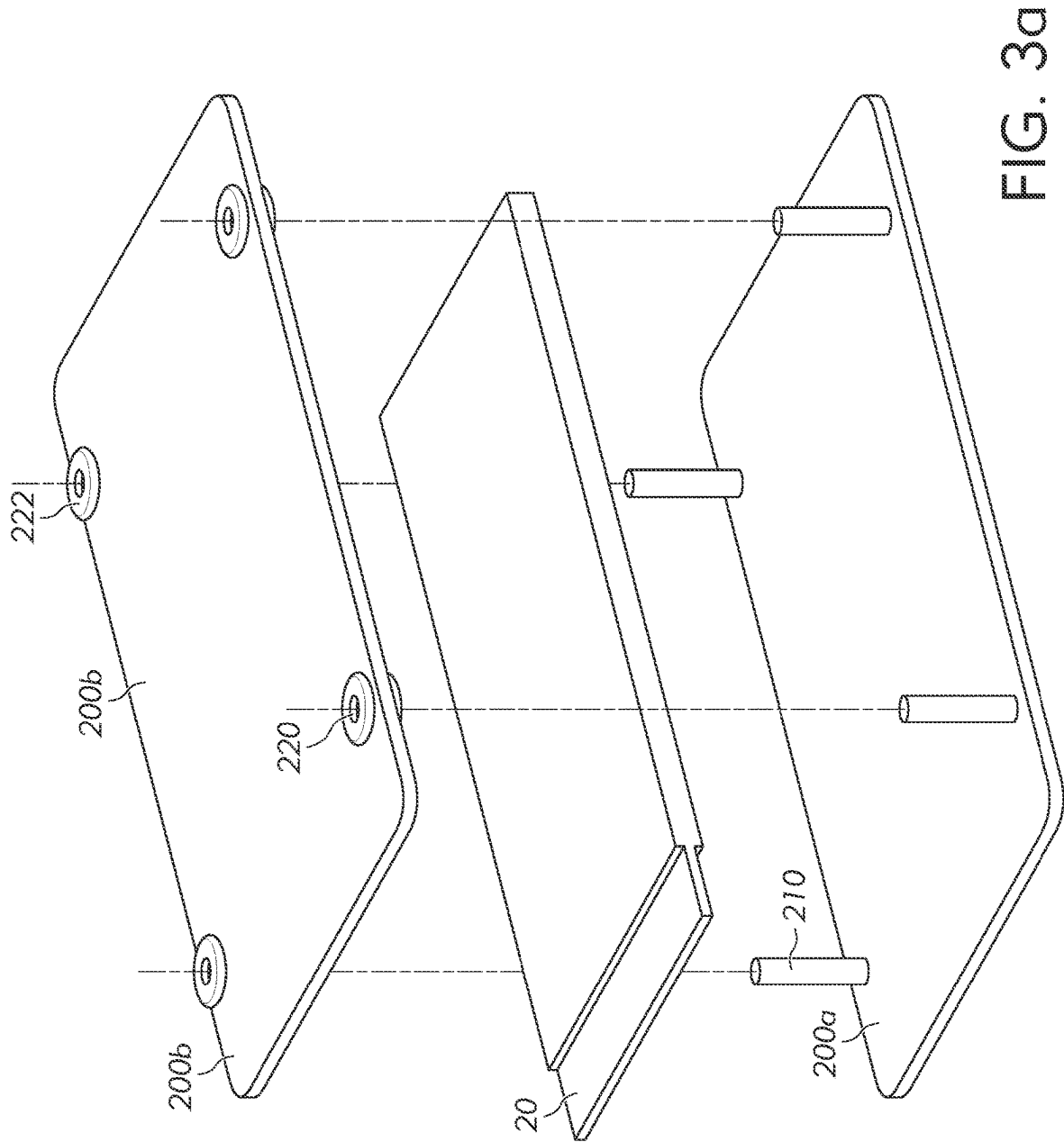


FIG. 3a

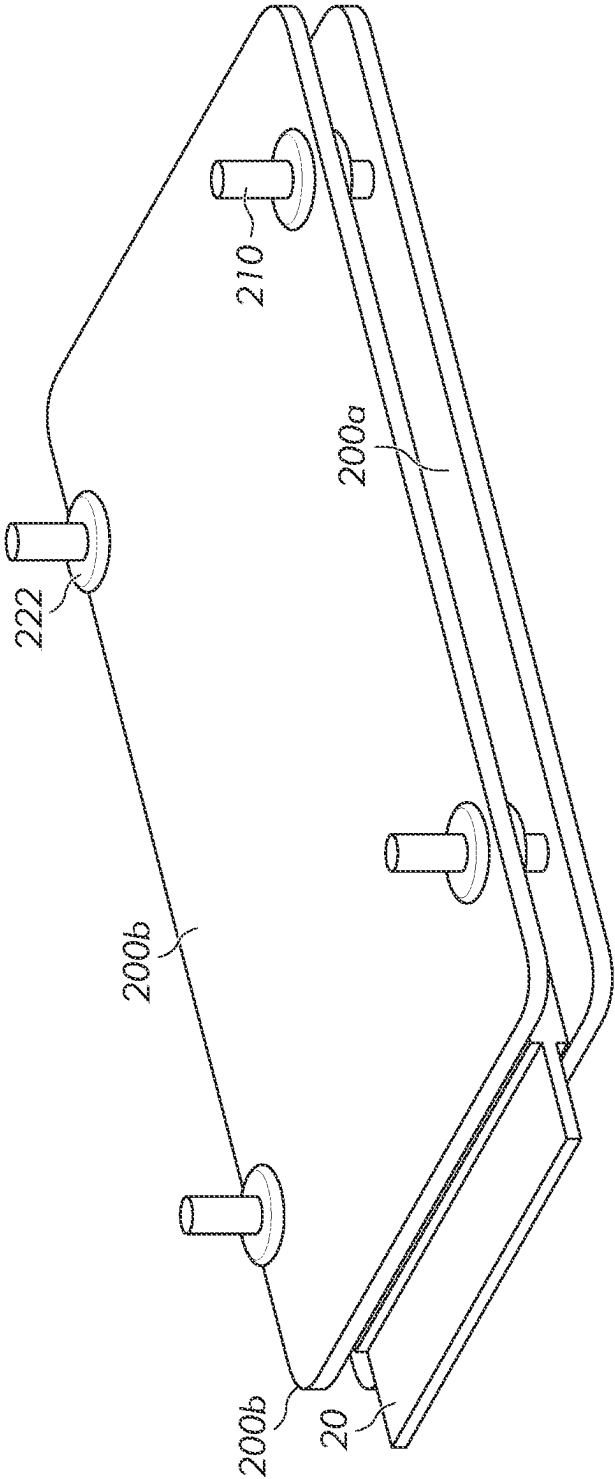


FIG. 3b

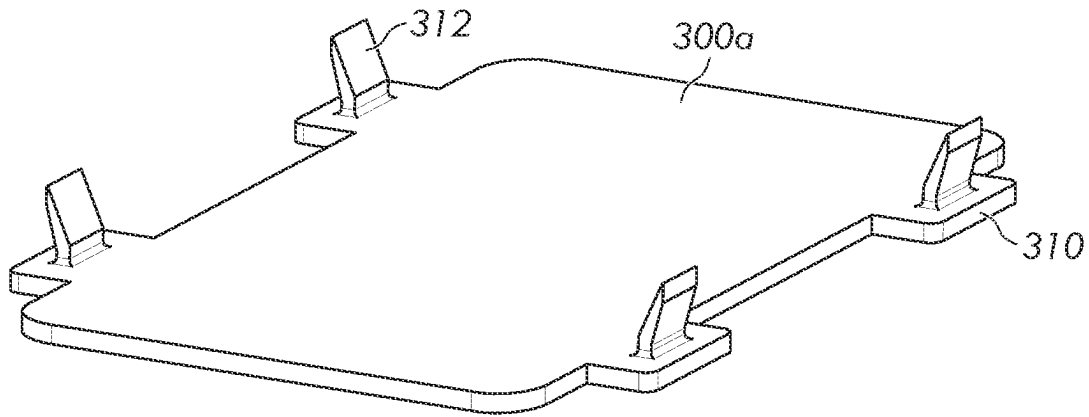


FIG. 4a

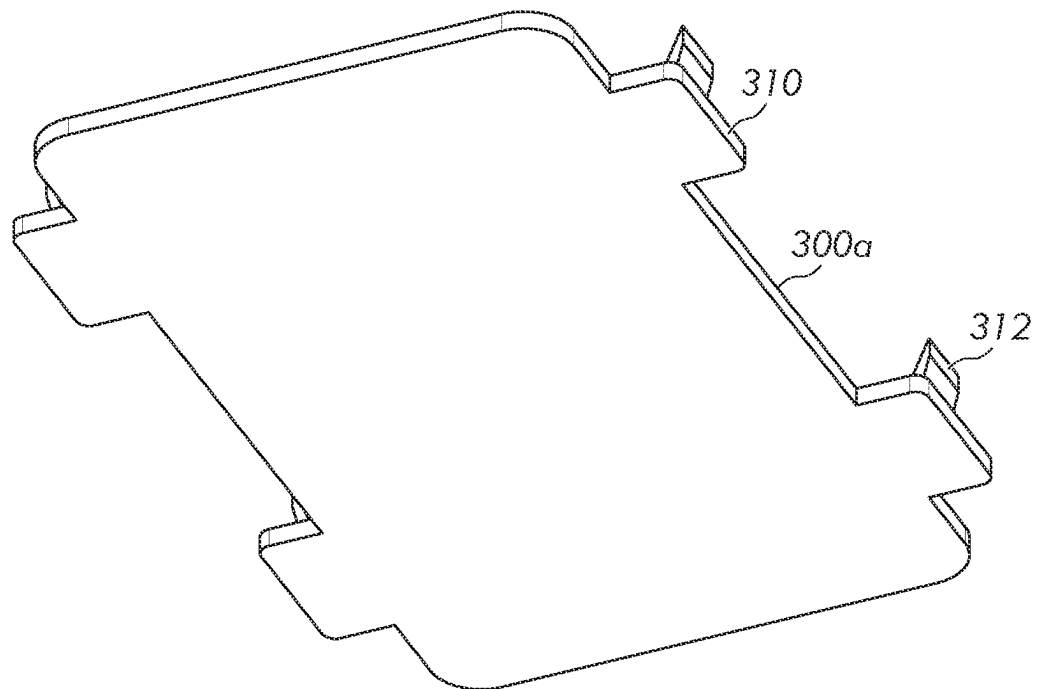


FIG. 4b

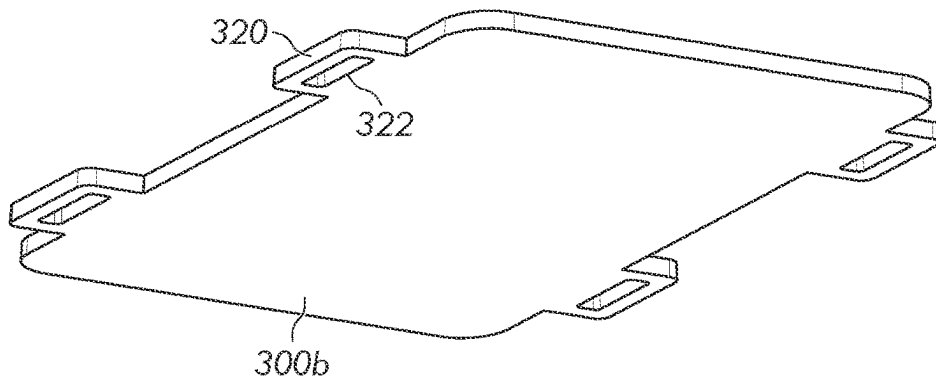


FIG. 4c

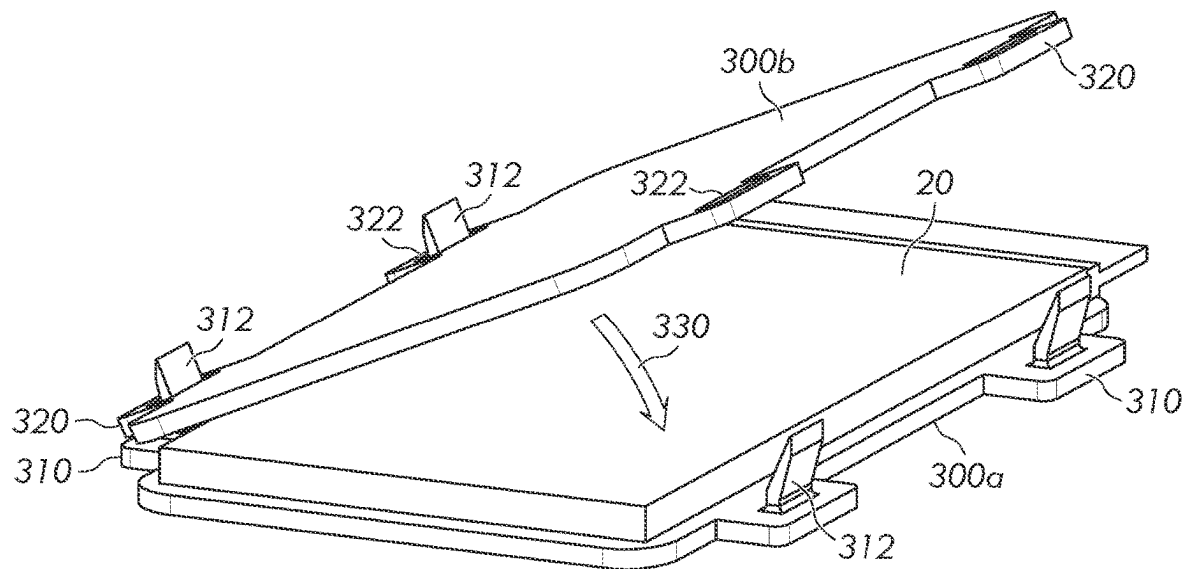


FIG. 4d

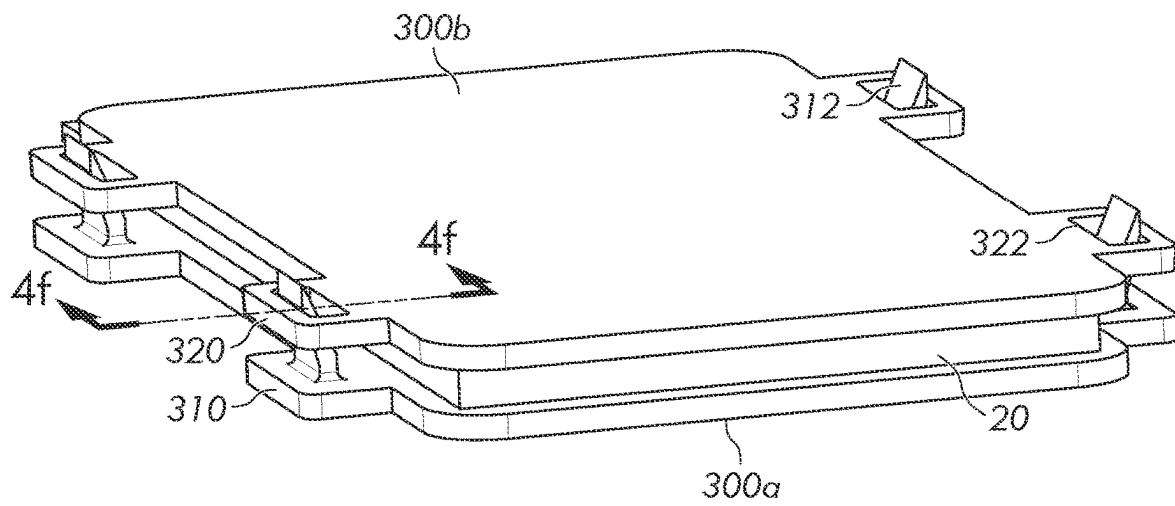


FIG. 4e

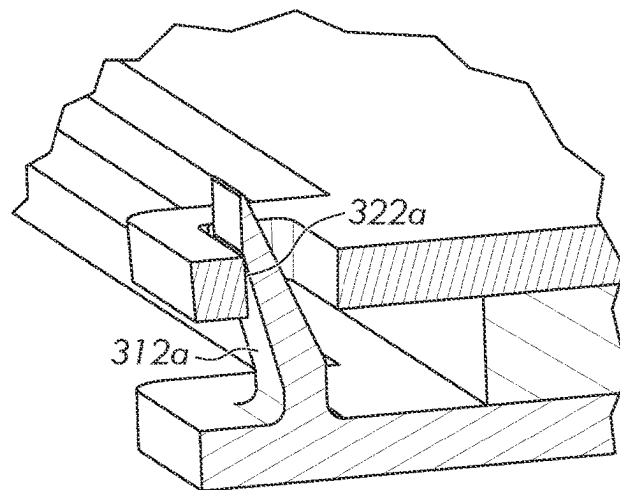
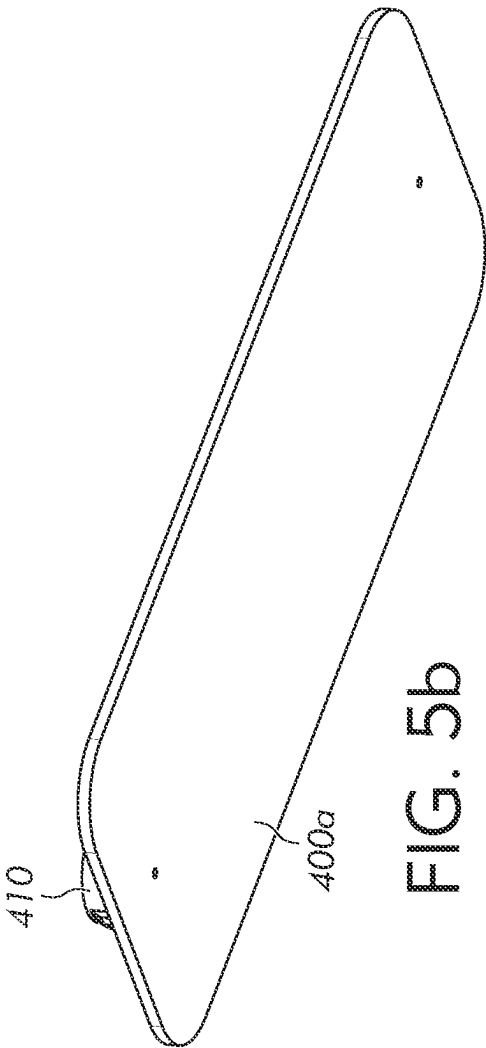
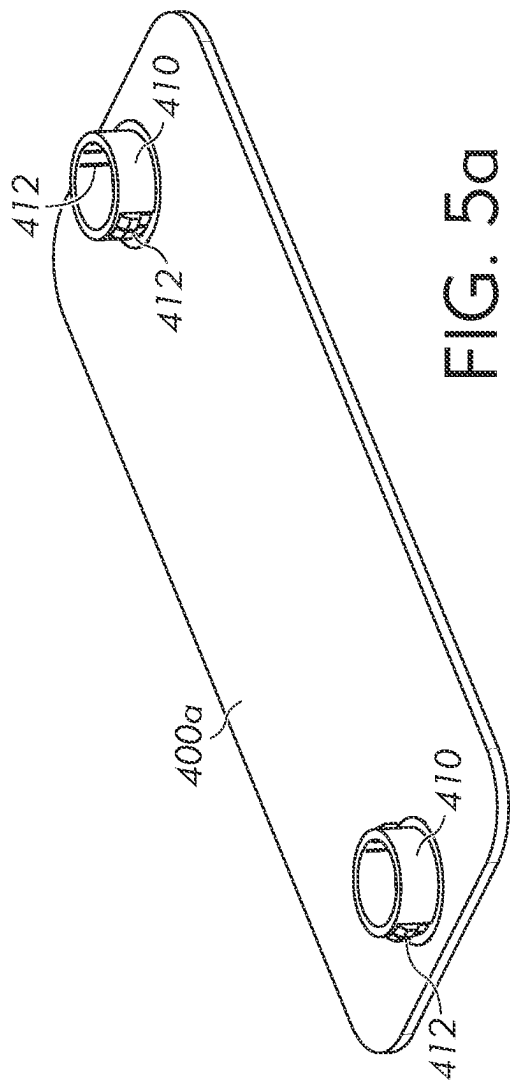
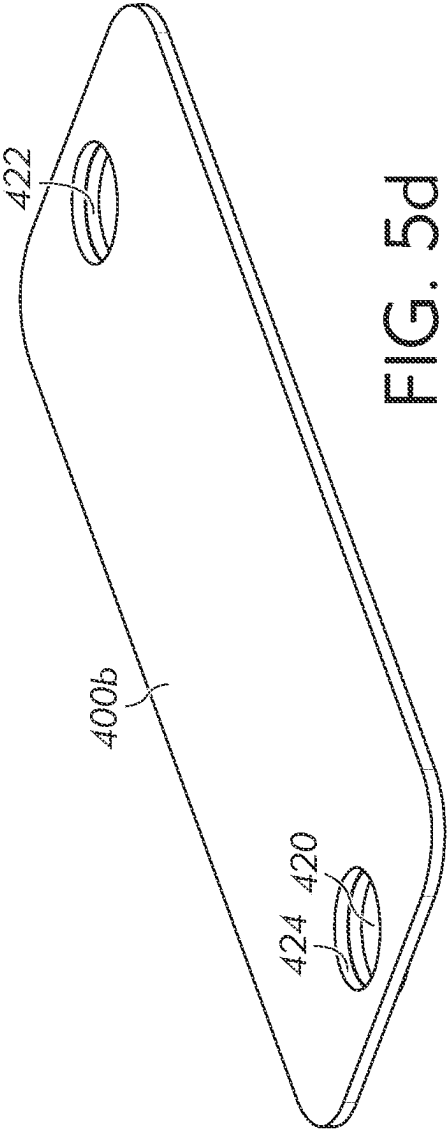
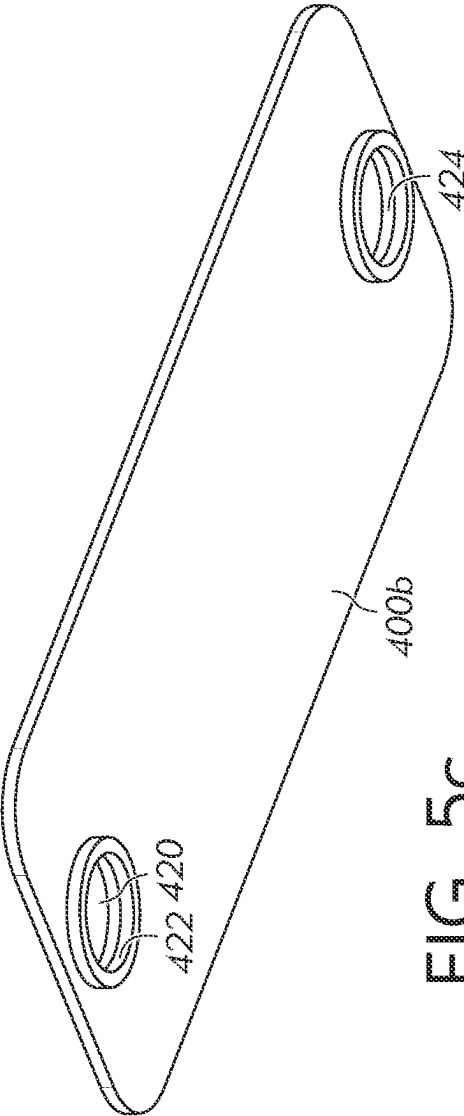


FIG. 4f





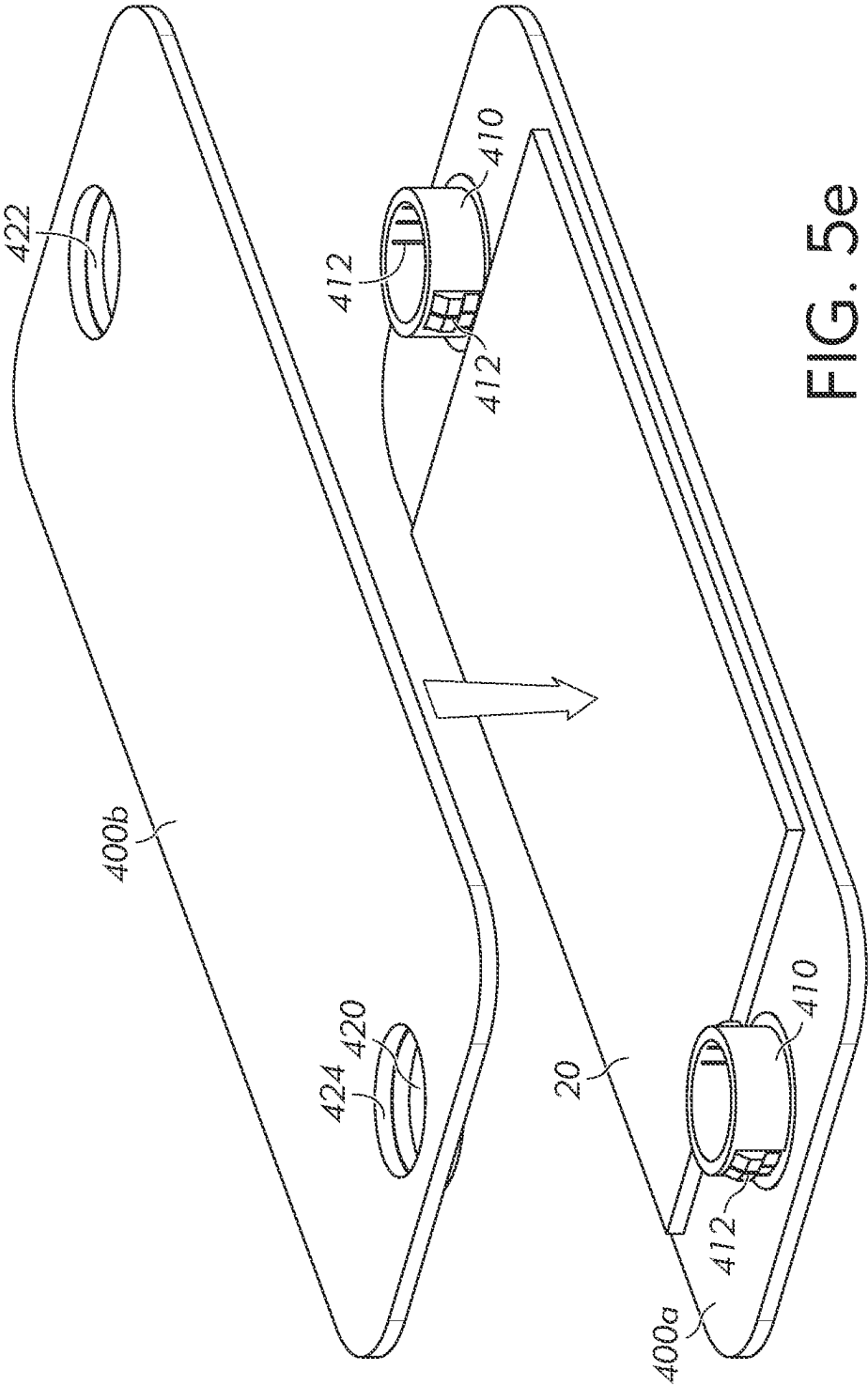
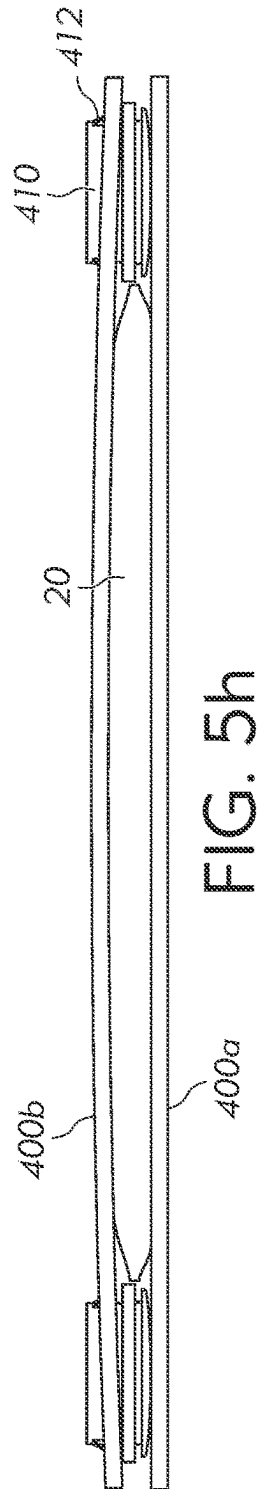
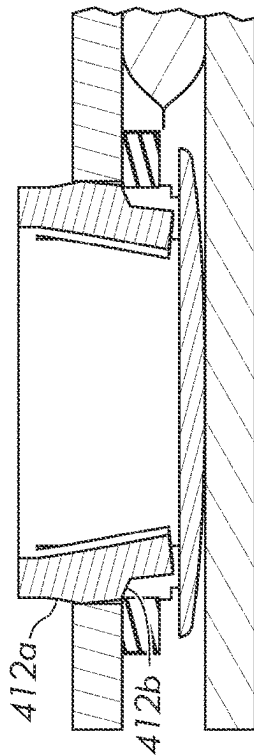
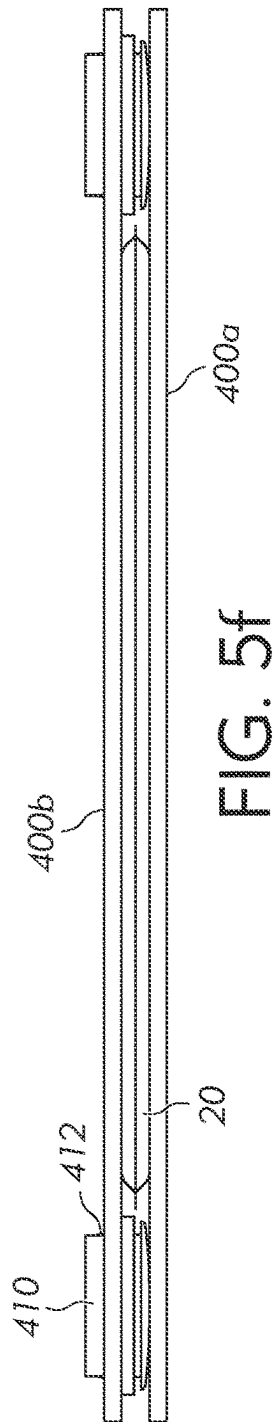


FIG. 5e



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DEVICE AND METHODS FOR FREEZING LIQUIDS INTO A FLAT FORM

FIELD OF THE INVENTION

The present invention relates in general to the storage of liquid-containing food and beverages and, more particularly, to devices and methods for freezing liquid-containing food and beverages into a flat form.

BACKGROUND OF THE INVENTION

Freezing potable or edible liquids is a good way to preserve the liquids for an extended period of time. In many cases, using a resealable plastic zipper bag is a cheap and convenient storage medium for freezing liquids. However, plastic zipper bags do not keep their shape well when filled with a liquid. FIG. 1*a* shows a plastic zipper bag **10** filled with a liquid. Bag **10** has an irregular shape and, once frozen, will keep that shape until thawed. The irregular shape of bag **10** when frozen solid makes the bag inconvenient to store. Especially when several of the bags are being used, the bags in combination end up taking up a lot more space in the freezer than is necessary if, e.g., all of the liquid was frozen as one solid mass. Bag **10** also doesn't stack well and the bags tend to slide off of each other and quickly become disorganized.

Freezing liquid in a flat form can improve storage efficiency. FIG. 1*b* shows a plastic zipper bag **20** that was held flat while the liquid inside was frozen so that the frozen solid liquid is now stuck in a regular flat shape. Having a liquid-filled plastic bag frozen in a flat form improves storage efficiency because the bags can be neatly stacked or stored upright in a bin **30** as shown in FIG. 1*c*. The flat surfaces of the plastic bags **20** contact each other neatly without significant air gaps formed by irregular surface shapes.

Having frozen liquids in a flat form is convenient for storage, but keeping the liquid-filled bags flat while the liquid freezes into a solid is not easy or simple. Therefore, a need exists for improved methods and devices for freezing liquids into a flat form.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1*a*-1*c* illustrate freezing liquid in plastic bags for storage;

FIGS. 2*a*-2*i* illustrate a device to hold a plastic bag in a flat form while freezing a liquid contained in the bag;

FIGS. 3*a* and 3*b* illustrate a second embodiment;

FIGS. 4*a*-4*f* illustrate a third embodiment; and

FIGS. 5*a*-5*h* illustrate a fourth embodiment.

DETAILED DESCRIPTION OF THE DRAWINGS

The present invention is described in one or more embodiments in the following description with reference to the Figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings.

FIGS. 2*a*-2*i* illustrate a device usable to hold a liquid in a plastic bag flat while the liquid is being frozen into a solid.

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The liquid being frozen is commonly freshly pumped breast milk. However, the devices described below can be utilized with any liquid, gel, or other freezable substance. The bag contents being frozen can include both liquid and solid components, e.g., soup can be frozen for long term storage. The bag contents may not be intended for consumption, e.g., water can be frozen to form ice packs. Even non-potable or inedible liquids may be frozen for long term preservation or storage.

FIGS. 2*a* and 2*b* illustrate a bottom plate **100a** of the device, while FIGS. 2*c* and 2*d* illustrate top plate **100b**. Bottom plate and top plate are arbitrary names, and the device can be used with either plate on the top, or even with the plates positioned next to each other or in any other desired orientation. Bottom plate **100a** and top plate **100b** are injection molded from plastic with nearly identical shapes. In other embodiments, plates **100a** and **100b** are 3d-printed, machined, or formed using other methods. The material of plates **100a** and **100b** can be any suitable polymer or plastic, wood, metals such as aluminum or stainless steel, or any other suitable material.

Both plates **100a** and **100b** include a flat portion **110** that occupies most of the footprint of the plates. Flat portions **110** are rectangular with rounded corners, but any suitable shape can be used. Flat portions **110** include inner surfaces **112** and outer surfaces **114**. In use, a plastic bag is pressed between the inner surfaces **112** of the plates **100a** and **100b** while being frozen to keep the liquid contained in the bag flat. Inner surfaces **112** are typically, but not necessarily, kept completely smooth and flat to create as flat of a surface to the frozen liquid as possible. Outer surfaces **114** will commonly have instructions, branding, or other information or designs molded into or printed on the outer surfaces.

Both plates **100a** and **100b** are formed with cylindrical interconnect structures **120** extending out above outer surfaces **114**. Interconnect structures **120** are cylinders with openings **122** formed through the centers of the cylinders. Openings **122** and interconnect structures **120** are concentric cylinders with each other and extend perpendicularly to outer surface **114**. Interconnect structures **120** extend over outer surfaces **114** so that the interconnect structures do not interfere with inner surfaces **112** of plates **100** squeezing against a liquid-filled bag. In other embodiments, interconnect structures **120** also extend at least partially in the opposite direction, over inner surfaces **112**, to maintain an offset between plates **100a** and **100b**.

Bottom plate **100a** has banana plugs **130**, or another type of spring pin, disposed in openings **122** and extending vertically over inner surface **112**. Banana plugs **130** are permanently molded into openings **122** in one embodiment. In other embodiments, banana plugs **130** are removably snapped or screwed into openings **122**. FIG. 2*e* shows a detailed view of a banana plug **130**. Banana plug **130** includes a compressible spring shaft **132** that extends over inner surface **112**. Spring shaft **132** is sized such that the banana plug shaft fits within openings **122** of top plate **100b**, although doing so requires the user to press the plates with enough force to compress the spring shaft inwards. The compression when spring shaft **132** of bottom plate **100a** is inserted into openings **122** of top plate **100b** applies pressure and helps hold the two plates together during the freezing process. The friction of banana plug **130** within opening **122** latches plate **100a** to plate **100b**.

Banana plugs **130** also include a hilt **134**. Hilt **134** is a flat plate oriented parallel to inner surface **112**. Hilt **134** presses against inner surface **112** when banana plug **130** is installed to help the banana plug be properly positioned by the user

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or factory manufacturing the bottom plate **100a**. Threads **136** are formed on the bottom of banana plug **130**. Threads **136** are helical to allow the banana plug to be screwed into and out of openings **122**. In other embodiments, threads **130** are replaced by one or more flanges or another type of structure to keep banana plugs **130** held within openings **122** when the banana plug threads are molded into bottom plate **100a**.

Four interconnect structures **120** with openings **122** are formed symmetrically near the middle of plates **100a** and **100b**. In other embodiments, interconnect structures **120** are formed in the corners of plates **100a** and **100b**. Some embodiments only use two interconnect structures **120** per plate, which could be centered along either the long or short edges of plates **100a** and **100b** or positioned in any other suitable locations. Any suitable number and positioning of interconnect structures **120** can be used. Forming interconnect structures **120** in symmetrical positions allows the same mold to be used for both plates **100a** and **100b**, while still having the interconnect structures align to each other when one of the plates is flipped.

Plates **100a** and **100b** are molded with finger grips **140** that extend vertically above outer surfaces **114**. Finger grips **140** provide a hollow space between plates **100a** and **100b** when the plates are pressed together. Depending on the thickness of a bag disposed between plates **100a** and **100b**, which will define a distance between the plates, a user may have trouble getting a finger between the plates to separate them. Finger grips **140** ensure that at any separation distance, a user can get a finger between plates **100a** and **100b**. Finger grips **140** include three perpendicularly connected sidewalls and a top. Finger grips **140** open out away from plate **100**. The top of finger grip **140** has a concave shape to be more ergonomic but could also align to the outer perimeter of flat portions **110**.

FIGS. **2f** and **2g** show usage of plates **100a** and **100b** to freeze a liquid into a flat form. In FIG. **2f**, a bag **20** filled with a freezable liquid is disposed between plates **100a** and **100b**. One convenient strategy is to place bag **20** on bottom plate **100a** first, and then clamp down top plate **100b** onto the bottom plate over the bag. This ensures that bag **20** is completely within banana plugs **130** and won't interfere with the banana plugs being inserted into openings **122**. Plates **100a** and **100b** are formed nearly identically, with interconnect structures **120** being located symmetrically. That way, when the plates are moved toward each other with inner surfaces **112** oriented toward each other, banana plugs **130** of bottom plate **100a** are aligned with and are able to extend into openings **122** of top plate **100b**.

FIG. **2g** shows plates **100a** and **100b** clamped, snapped, pressed, or latched together with banana plugs **130** of bottom plate **100a** inserted into openings **122** of top plates **100b**. Each spring shaft **132** is compressed inward by being forced into an opening **122** and applies a force outward against interconnect structures **120** accordingly. The force being applied by banana plugs **130** keeps plates **100a** and **100b** attached to each other during the freezing process. Banana plugs **130** remain slidable in and out of openings **122**, which allows plates **100a** and **100b** to separate slightly as the liquid freezes and expands without damaging the plates. The same mechanism allows the user to pull plates **100a** and **100b** apart relatively easily once the liquid in bag **20** has frozen. Finger grips **114** ensure that there is sufficient space between plates **100a** and **100b** for a user to insert a finger and get a sufficient grip to pull the plates apart.

FIGS. **2h** and **2i** illustrate bag **20** after having been frozen. Bag **20** keeps its shape even though there are no more plates

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100 supporting the bag's shape. The liquid is frozen solid and maintains its shape. The liquid can now be stored in a convenient manner that is easy to organize.

FIGS. **3a** and **3b** illustrate a similar embodiment with bottom plate **200a** and top plate **200b**. Instead of having banana plugs **130** that press outward against an opening, bottom plate **200a** has fixed pegs **210** and top plate **200b** has openings **220** with rubber gromets **222** that press inward to grip the fixed pegs. Bag **20** is inserted between bottom plate **200a** and top plate **200b** as shown in FIG. **3a**, and then pegs **210** are pressed into openings **220** as shown in FIG. **3b**. The rubber gromets **222** are compressed between the material of top plates **200b** and pegs **210**, creating friction to keep the plates attached to each other. Plates **200** are formed from any of the materials and using any of the methods discussed above for plates **100**. In one embodiment, plates **200** are cut from sheet metal.

Pegs **210** and openings **220** are formed at symmetrical or the same positions on their respective plates so that they can be easily aligned to each other. Plates **200a** and **200b** have substantially the same shape as plates **100a** and **100b** and can be formed from similar materials and in similar manufacturing methods. Pegs **210** can be formed as part of plate **200a** or formed separately and attached by welding, being molded with plate **200a**, with a screw or bolt, using an adhesive, or using any other suitable method. Plates **200a** and **200b** can be formed with finger grips **140** as shown above.

FIGS. **4a-4f** illustrate a third embodiment with clips used to hold two plates **300a** and **300b** together. FIGS. **4a** and **4b** show bottom plate **300a** and FIG. **4c** shows top plate **300b**. Plates **300** are formed of any of the materials discussed above for plates **100** and using any of the methods mentioned. In one embodiment, plates **300** are injection molded plastic. Finger grips **140** are added in some embodiments.

Bottom plate **300a** has four tabs **310** extending outward from the plate, within the same plane, to allow clips **312** to rise vertically outside the main rectangle of the plates. Clips **312** extend upward and outward from tabs **310** so that a distance between two opposing clips at the top of the clips is greater than a distance between the same clips at their base. Top plate **300b** has four tabs **320** that extend outward from the main body of the top plate in similar locations and with similar sizes to tabs **310** of bottom plate **300a**. Tabs **310** have openings or slots **322** formed through the tabs.

The four slots **322** are configured to allow all four clips **312** to extend through the slots and thereby latch top plate **300b** onto bottom plate **300a**. Clips **312** slant outward and need to be pressed inward toward the main body of bottom plate **300a** to fit all four clips through their respective slots **322**. Once clips **312** are inserted through slots **322**, the slanted outer surfaces of the clips press against surfaces within slots **322** to keep plates **300a** and **300b** latched together.

Inserting all four clips **312** through the four slots **322** at once can be difficult because the clips need to be pressed inward while aligning all four clips to all four opening at once. FIG. **4d** illustrates one more convenient method of clipping top plate **300b** to bottom plate **300a**. First, two clips on one side of bottom plate **300a** are inserted through two slots **322** on the same side of top plate **300b**, which requires no bending of any clips **312** because the two clips on the same side of the plate or slanted in the same direction. Then, top plate **300b** is rotated down as indicated by arrow **330** until the opposing clips **312** on the other side are physically touching their respective tabs **320**. Only a small force is required by the user to slide the remaining two openings **322**

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laterally and over the two remaining clips **312** because clips **312** on the first side already being inserted through slots **322** keep the plates properly aligned through the process.

FIGS. **4e** and **4f** show clips **312** engaged with slots **322**. A bag **20** is squeezed between plates **300a** and **300b** based on the force of clips **312** being applied against top plate **300b**. FIG. **4f** is a zoomed in cross-sectional view showing details of the sloped outer surface **312a** of tab **312** pressing against surface **322a** within opening **322**. In some embodiments, inner surface **322a** is beveled or sloped to make it easier to press clips **312** through slots **322**. All four clips **312** can then be inserted partially into slots **322** without having to bend the clips, and then pressure from plates **300a** and **300b** being pressed together can stretch clips **312** inward as the clips slide along the sloped surfaces **322a**.

Once the liquid within bag **20** is frozen, plates **300a** and **300b** are separated by pulling them apart. Clips **312** again bend inward as the sloped surface **312a** slides against the inner surface of slots **322** and out of the slots. The frozen flat bag **20** can be conveniently stored.

FIGS. **5a-5h** illustrate another embodiment with a different style of clip. FIGS. **5a** and **5b** show bottom plate **400a**. Bottom plate **400a** has two interconnect structures **410**, one at each end of the bottom plate. Interconnect structures **410** have a circular shape with clips **412** formed on two opposing sides of each interconnect structure. Clips **412** extend outward from the circular shape of interconnect structure **410** and are compressible inward such that the outer surfaces of clips **412** can be pressed to within the circle of the interconnect structure when fully compressed inward. Clips **412** spring back outward when the compression is released.

FIGS. **5c** and **5d** show top plates **400b**. Top plate **400b** has two openings **420** formed near the ends of the top plate in locations corresponding to interconnect structures **410** on bottom plate **400a**. Openings **420** are positioned to be aligned to interconnect structures **410** when top plate **400b** is placed over bottom plate **400a** as shown in FIG. **5e**. When plates **400** are moved together, interconnect structures **410** are inserted into openings **420** as shown in FIGS. **5f** and **5g**.

Openings **420** have a narrow portion **422** and a wider portion **424**. Interconnect structures **410** are inserted with narrow portion **422** oriented toward bottom plate **400a**. Narrow portion **422** has a diameter that is the same or slightly larger than a diameter of interconnect structures **410**, but less than the distance between the outermost portions of clips **412** when the clips are not pressed in. Therefore, as plates **400** are pressed together around bag **20** filled with a freezable liquid, narrower portion **422** of opening **420** fits around interconnect structure **410** and presses clips **412** inward.

Clips **412** have sloped upper surfaces **412a**, better illustrated in the zoomed-in cross-sectional view of FIG. **5g**, that allow the surface of narrower portion **422** to slide down while pressing clips **412** inward gradually. Once plates **400a** and **400b** have been pressed together sufficiently far that the wider portion of clip **412** is completely through narrower portion **422** of opening **420**, the clips expand outward into wider portion **424** of opening **420**. Clips **412** are not able to completely expand back outward within opening **420**, so a force is applied by the clips to the inside of the opening. The force being applied by clips **412** helps keep plates **400** together and keeps clips **412** from sliding back over narrower portion **422** of opening **420**. The plates press against bag **20** to keep the liquid contained therein in a flat form.

Any pressure pulling plates **400** apart will have to be sufficient enough to overcome the spring force of clips **412**. Clips **412** have a bottom sloped surface **412b** that will press

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against narrower portion **422** of opening **420**. Sloped surface **412b** allows the narrower portion **422** to compress clips **412** inward when a user applies sufficient force to pull plates **400** apart.

Plates **400a** and **400b** are made of flexible plastic so that, as the liquid in bag **20** freezes and expands, the plates are able to expand outward while clips **412a** remain engaged with openings **420**. The flexible aspect of plates **400a** and **400b** also allows a thicker or over-filled bag **20** to be pressed between the plates. FIG. **5h** illustrates a bag **20** with more liquid than in the bags illustrated above being frozen with plates **400**. The plates **400** are able to flex outward to accommodate the thicker bag **20**. Because interconnect structures **410** are formed near the ends of plates **400** along their longest dimension, plates **400** have more length to flex. In other embodiments, plates **400** are formed using any of the methods and materials discussed above for plates **100**.

In each embodiment, two plates are used to keep a bag of liquid pressed between the plates in a flat form while the liquid freezes. The plates have a two-part latching mechanism, where one part of the latching mechanism is on one plate and a second part of the latching mechanism on the other plate. As the plates are moved together, the latching mechanism latches the two plates together. The latching mechanism can be a banana plug into an opening, a peg into a grommet, a slanted clip into a slot, a round clip into a round opening, or any other suitable latching mechanism. Any of the above embodiments can have their respective latching mechanism formed at the ends of their respective plates to allow the plates to flex.

While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims. Those having ordinary skill in the art will recognize that the disclosed features can be used in different combinations than those specifically disclosed when the features are compatible with each other.

What is claimed is:

1. A method of freezing liquid in a flat form, comprising: providing a first plate including,

a first surface, wherein the first surface is completely flat and extends to four outer edges of the first plate, and

a spring pin extending over the first surface;

providing a second plate including,

a second surface, wherein the second surface is completely flat and extends to four outer edges of the second plate, and

a cylindrical opening formed through the second plate; disposing a bag containing a freezable liquid between the first plate and second plate with the first surface of the first plate and the second surface of the second plate oriented toward the bag; and

inserting the spring pin of the first plate into the cylindrical opening of the second plate to attach the first plate to the second plate with a bag containing freezable liquid squeezed between the first surface of the first plate and the second surface of the second plate, wherein the spring pin applies a force outward against a cylindrical inner surface of the cylindrical opening while remaining slidable in the cylindrical opening to allow a separation between the first plate and second plate to increase as the freezable liquid freezes and presses outwardly against the first plate and second plate.

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2. The method of claim 1, wherein the protrusion includes a spring pin includes a banana plug.

3. The method of claim 1, further including placing the first plate, second plate, and bag into a freezer.

4. The method of claim 1, further including screwing the spring pin into the first plate. 5

5. A method of freezing a liquid in a flat form, comprising: providing a first plate including a spring pin; providing a second plate including a cylindrical opening; and 10

inserting the spring pin into the cylindrical opening to couple the first plate to the second plate, wherein the spring pin applies a force outward against a cylindrical inner surface of the cylindrical opening while remaining slidable in the cylindrical opening to allow a separation between the first plate and second plate to increase as the liquid freezes and presses outwardly against the first plate and second plate. 15

6. The method of claim 5, wherein the spring pin includes a banana plug. 20

7. The method of claim 5, further including molding the spring pin into the first plate.

8. The method of claim 5, further including screwing the spring pin into the first plate. 25

9. The method of claim 5, further including: disposing a bag containing the liquid between the first plate and second plate; and placing the first plate, second plate, and bag into a freezer. 30

10. The method of claim 9, wherein the first plate and second plate press against the bag to keep the liquid in a flat form. 35

11. The method of claim 5, wherein the first plate includes a plurality of symmetrically located spring pins and the second plate includes a plurality of symmetrically located cylindrical openings.

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12. The method of claim 5, wherein:

the first plate includes a first surface that is completely flat for an entire footprint of the first surface;

the second plate includes a second surface that is completely flat for an entire footprint of the second surface; and

the liquid is squeezed between the first surface of the first plate and the second surface of the second plate.

13. A method of freezing liquid in a flat form, comprising: providing a first plate including a spring pin; providing a second plate including a cylindrical opening extending completely and continuously through the second plate; and

inserting the spring pin into the cylindrical opening to attach the first plate to the second plate.

14. The method of claim 13, wherein the spring pin includes a banana plug.

15. The method of claim 13, further including molding the spring pin into the first plate.

16. The method of claim 13, further including removably attaching the spring pin to the first plate.

17. The method of claim 13, wherein the first plate includes a plurality of symmetrically located spring pins.

18. The method of claim 13, wherein the spring pin applies a force outward against a cylindrical inner surface of the cylindrical opening while remaining slidable in the cylindrical opening.

19. The method of claim 13, wherein:

the first plate includes a first surface that is flat for an entire footprint of the first surface; and

the second plate includes a second surface that is flat for an entire footprint of the second surface.

20. The method of claim 19, further including:

disposing the liquid in a bag; and squeezing the bag between the first surface and the second surface.

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